Run-down Correction Procedure

Problem

During repeated activations of skinned cardiac muscle, progressive force loss (run-down) occurs due to cumulative damage and ATP depletion. A reference recording `F_ref` and a later run-down recording `F_rd` show a systematic force deficit that varies with sarcomere length (SL) and time within the activation. The goal is to reconstruct what `F_rd` would have looked like without run-down, and to correct `F_ref` for its own within-recording decay.

Data

Recording	Dataset	Description
F_ref	data(2)	8 mM ATP, active — first (reference) activation
F_rd	data(4)	8 mM ATP, active — repeat activation (impaired)

Both recordings follow the same staircase protocol (SL = 2.0 um baseline, steps to 2.2 um and back) over a ~10 s window (wall-clock t = 68 – 78 s). Force is measured in kPa.

Two stable isometric zones at SL = 2.0 um are used for calibration:

- **Zone 1**: t = 71.5 – 72.4 s (reference time point)
- **Zone 2**: t = 77.4 – 78.4 s

Measured quantities

Quantity	Value	Source
F_ref at zone 1	68.3 kPa	

Quantity	Value	Source
		Linear fit to zone 1+2 points of data(2)
F_rd at zone 1	56.4 kPa	Linear fit to zone 1+2 points of data(4)
slope_ref	-0.495 kPa/s	Within-recording force decay of F_ref
slope_rd	-0.467 kPa/s	Within-recording force decay of F_rd
Force gap at zone 1	11.9 kPa	F_ref - F_rd at zone 1 midpoint

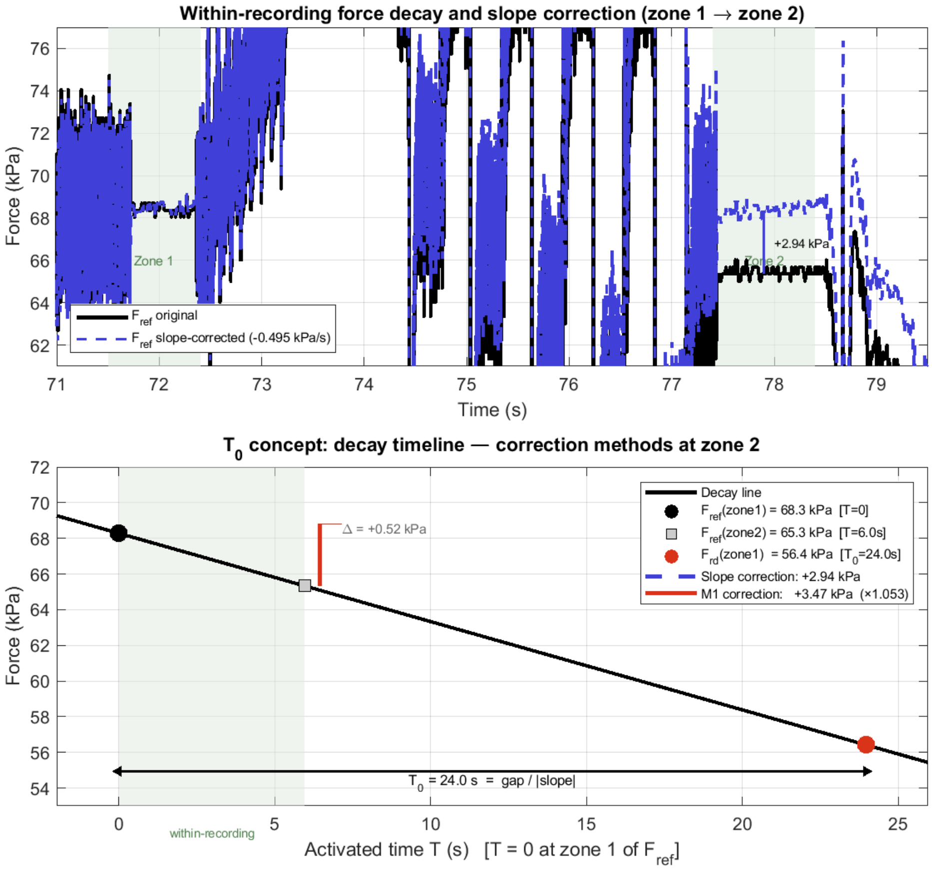
T0: hypothetical decay distance

The key derived quantity is **T0** — the hypothetical activated time it would take for force to decay from the reference level to the run-down level at a constant rate:

```
T0 = (F_ref(zone1) - F_rd(zone1)) / |slope_ref| = 11.9 / 0.495 = 24.0 s
```

T0 is not wall-clock time between recordings. It represents **equivalent activated time**: the cumulative hot-activated exposure (muscle under calcium) that separates the two recordings. Inactive recovery periods between activations do not contribute to T0.

T0 establishes a common time axis for the decay model: zone 1 of F_ref is at T = 0, zone 1 of F_rd is at T = T0.



T0 concept and within-recording decay

Top panel: Zoomed timecourse (t = 71–79.5 s) showing F_ref original (black) and slope-corrected (blue dashed). Zone 1 (t = 71.5–72.4 s) and Zone 2 (t = 77.4–78.4 s) are shaded. The slope correction (+2.94 kPa additive, SL-independent) is shown between the two zones. Bottom panel: Decay timeline concept. The dashed line shows force decay from zone 1 of F_ref (T = 0) at rate slope_ref. Zone 1 of F_rd falls at T = T0 = 24.0 s. Blue dashed arrow: slope correction at zone 2. Red arrow: M1 model correction at zone 2.

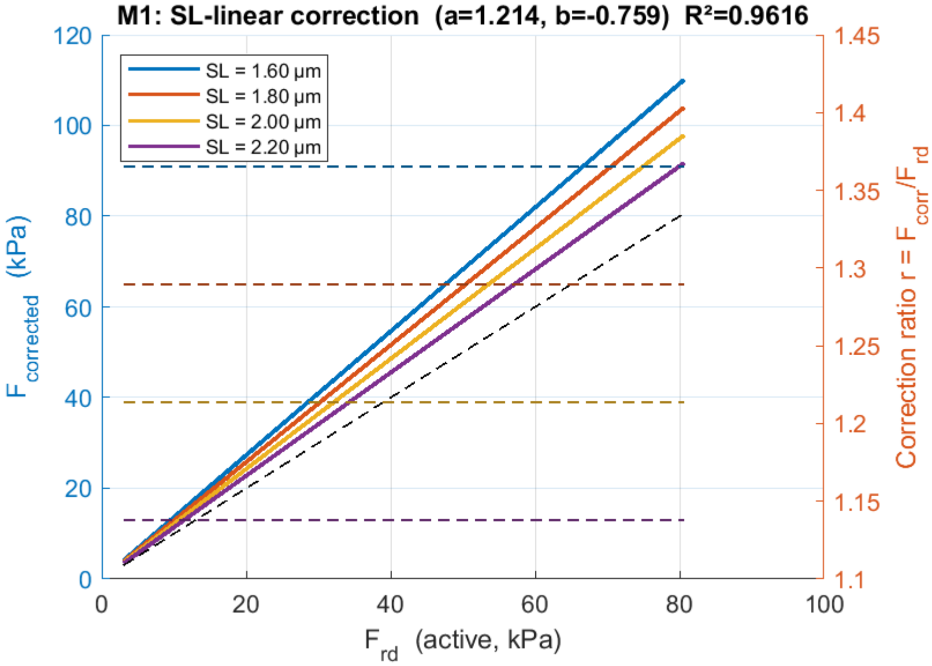
Step 1: Spatial correction model (M1) and F_rd correction

M1 model

A linear correction surface is fitted from paired (F_rd, F_ref) data over the valid window (F > 3 kPa, t = 68–78 s). Both signals are first slope-detrended to their zone-1 values before fitting, to remove within-recording trends. F_ref is processed on the hi-fi smoothed grid (`t_fineShifted`); F_rd is processed on the original measured points (`t_shifted_orig`), preserving the raw signal without smoothing.

Model M1 (SL-linear, 2 parameters):

```
F_corrected = F_rd * (a + b * (L - L0))    where L = SL/2, L0 =
1.0

a = 1.214,  b = -0.759

R^2 = 0.962,  RMSE = 2.1 kPa  (999 fitting points)
```



M1 correction curves at 4 SL levels

M1 correction curves at four sarcomere lengths. Left axis: corrected force vs run-down force. Right axis: correction ratio r = F_corr/F_rd. Dashed horizontal lines indicate the mean corrected force at each SL level. Higher correction at shorter SL, lower at longer SL — consistent with run-down impairing force more at shorter lengths.

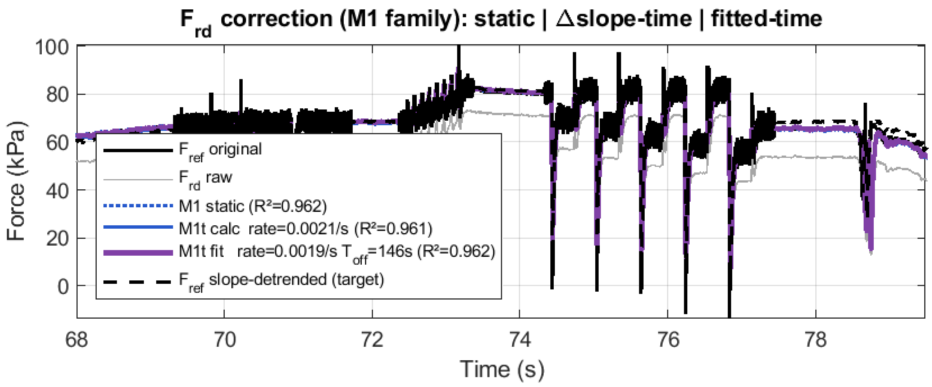
The correction ratio `a + b*(L - L0)` depends only on sarcomere length:

SL (um)	L = SL/2	Correction ratio
1.6	0.8	1.366
1.8	0.9	1.290
2.0	1.0	1.214
2.2	1.1	1.138

A more complex model (M3: +F×L interaction, 4 params, R^2=0.963) is available in the code but M1 is sufficient for practical data processing.

F_rd correction timecourse

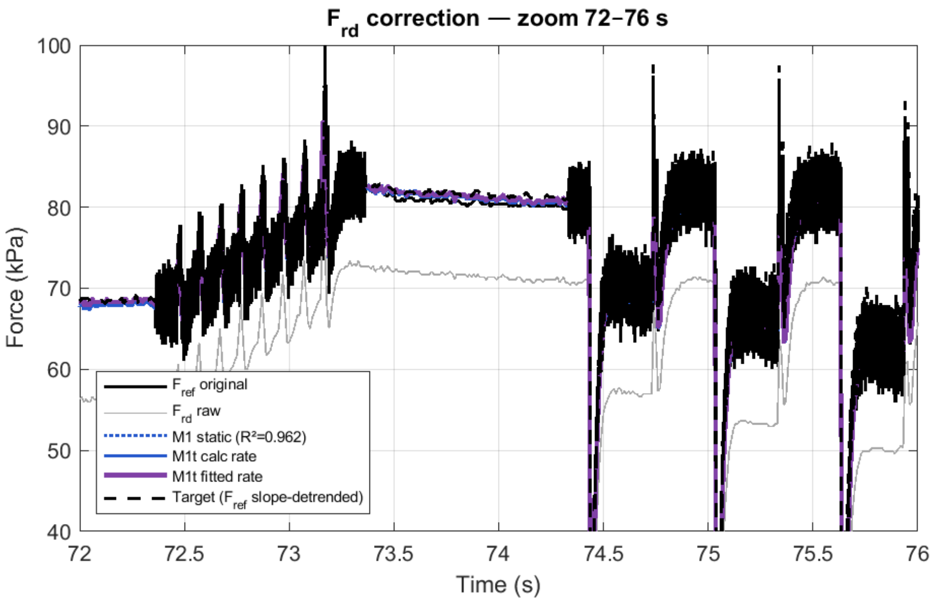
Applying the M1 correction brings the run-down recording back up to the slope-detrended reference level (dashed goal line).



F_rd correction timecourse

F_rd correction timecourse. Black: F_ref original. Grey: F_rd raw. Dotted blue: M1 static correction. Solid blue: M1 + time correction (rate from slope difference). Purple: M1 + time (freely fitted rate). Dashed black: F_ref slope-detrended (target). The goal line is plotted last and sits on top.

The zoom below confirms that all M1 variants converge tightly on the target:



F_rd correction zoom 72–76 s

Zoom of the F_rd correction at t = 72–76 s, 40–100 kPa. Static M1 (dotted) and time-extended variants (solid) land essentially on the slope-detrended F_ref target (dashed) throughout the slack and isometric segments.

Step 2: Model-based F_ref correction

The reference recording also decays within its own activation window. Rather than correcting with a simple constant slope (-0.495 kPa/s), we reuse the spatial model M1 scaled by the fraction of T0:

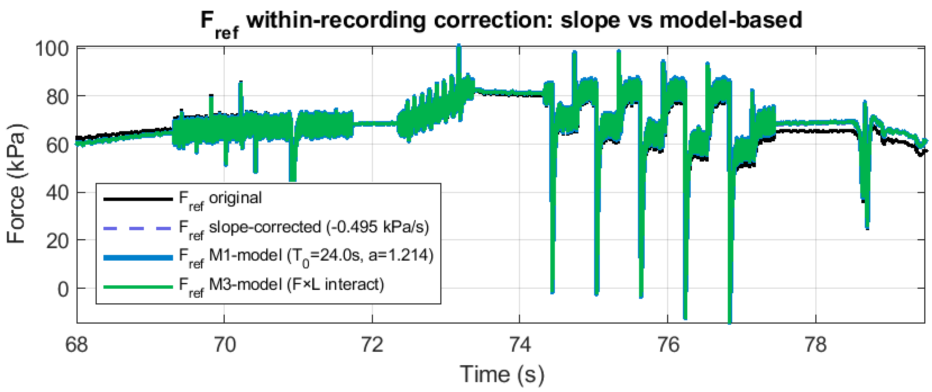
```
frac(t) = (t - t_zone_ref) / T0

r_model(t) = a + b * (L(t) - L0)      [same M1 correction ratio]

F_ref_corrected(t) = F_ref(t) * (1 + (r_model(t) - 1) * frac(t))
```

At zone 1 (frac = 0): no correction (identity).
At zone 2 (frac ≈ 0.25): correction of ~5.3% at SL = 2.0 (~+3.5 kPa on 68 kPa).
At T0 distance (frac = 1): full model correction (~21.4%).

Comparison: slope vs model at zone 2 (SL = 2.0)

Method	Correction	Effect on 68 kPa
None (original)	0	68.3 kPa → ~65.3 kPa at zone 2
Slope only	+2.94 kPa (additive)	65.3 + 2.9 = 68.3 kPa
M1 model	×1.053 (multiplicative)	65.3 × 1.053 = 68.7 kPa

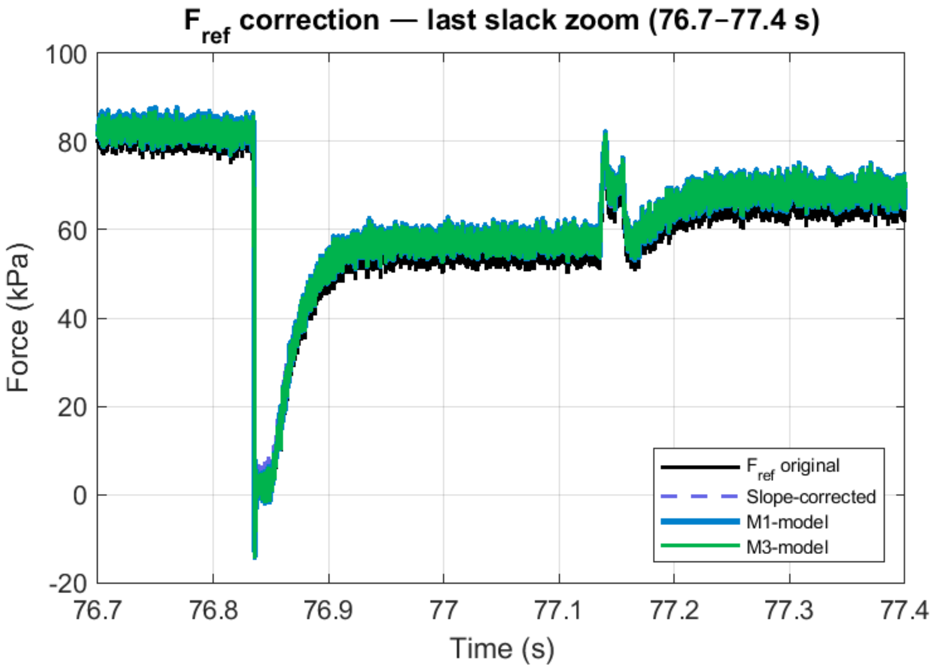
The model-based correction is **multiplicative and SL-dependent**: at SL = 2.2 the correction is smaller (~4.0%), at SL = 1.8 it is larger (~6.5%). The slope correction is additive and constant across all SL levels — a less accurate approximation of the underlying decay process.



F_ref correction timecourse

F_ref correction comparison over the full activation window. Black: original. Blue dashed: slope-corrected. Cyan: M1 model-based. Green: M3 model-based. All three correction methods diverge slightly at the SL = 2.2 stretch step where the M1/M3 corrections are smaller (as expected from the SL-dependence of run-down).

The last slack segment (76.7–77.4 s) provides the best view of the correction magnitude:



F_ref last slack zoom

Zoom of the last slack segment (SL ≈ 1.88 µm, t = 76.7–77.4 s). Slope correction (blue dashed) and M1/M3 model corrections (cyan/green) agree to within ~0.5 kPa. The correction lifts the steady-state isometric level by ~1.5 kPa relative to the uncorrected signal.

Output files

All saved as tab-delimited `[time, SL, force]` in the data directory:

File	Grid	Description
`Corr_Fref_original.txt`	hi-fi	F_ref as recorded (no correction)
`Corr_Fref_slopeOnly.txt`	hi-fi	F_ref with linear slope correction to zone 1
`Corr_Fref_M1_model.txt`	hi-fi	F_ref with M1 model-based correction to zone 1
`Corr_Fref_M3_model.txt`	hi-fi	F_ref with M3 model-based correction to zone 1

File	Grid	Description
`Fref_lofi.txt`	lo-fi	F_ref original, unsmoothed (t_shifted_orig)
`Corr_Frd_original.txt`	lo-fi	F_rd as recorded (no correction)
`Corr_Frd_M1_static.txt`	lo-fi	F_rd corrected by M1 static model
`Corr_Frd_M1t_slopeRate.txt`	lo-fi	F_rd corrected by M1 + time (rate from slope)
`Corr_Frd_M1t_fittedRate.txt`	lo-fi	F_rd corrected by M1 + time (rate freely fitted)
`Corr_Frd_M3_static.txt`	lo-fi	F_rd corrected by M3 static model

F_ref files are saved on the smoothed hi-fi grid (`t_fineShifted`); F_rd files are saved on the original measured points (`t_shifted_orig`).

How to apply to new data

For a new run-down recording with a known reference:

1. Fit the within-recording slope at two SL = 2.0 zones
2. Compute T0 = force_gap / |slope|
3. Apply M1 correction: `F_corrected = F_rd * (1.214 - 0.759 * (SL/2 - 1.0))`

For correcting F_ref itself:

1. Compute `frac = (t - t_zone1_midpoint) / T0`
2. Apply `F_ref_corrected = F_ref * (1 + (1.214 - 0.759*(SL/2-1.0) - 1) * frac)`

For multiple sequential recordings at known activation intervals: each recording is at a different T on the decay timeline. The same model and T0 concept extends naturally — the correction at each recording scales with its T distance from the reference.

Script

All fitting, visualization, and export is in `Workbench/FitRundownCorrection.m`. The script is self-contained (loads data from `AllDataMerged.mat`).

Feature analysis: effect of WRR correction on ktr and slack

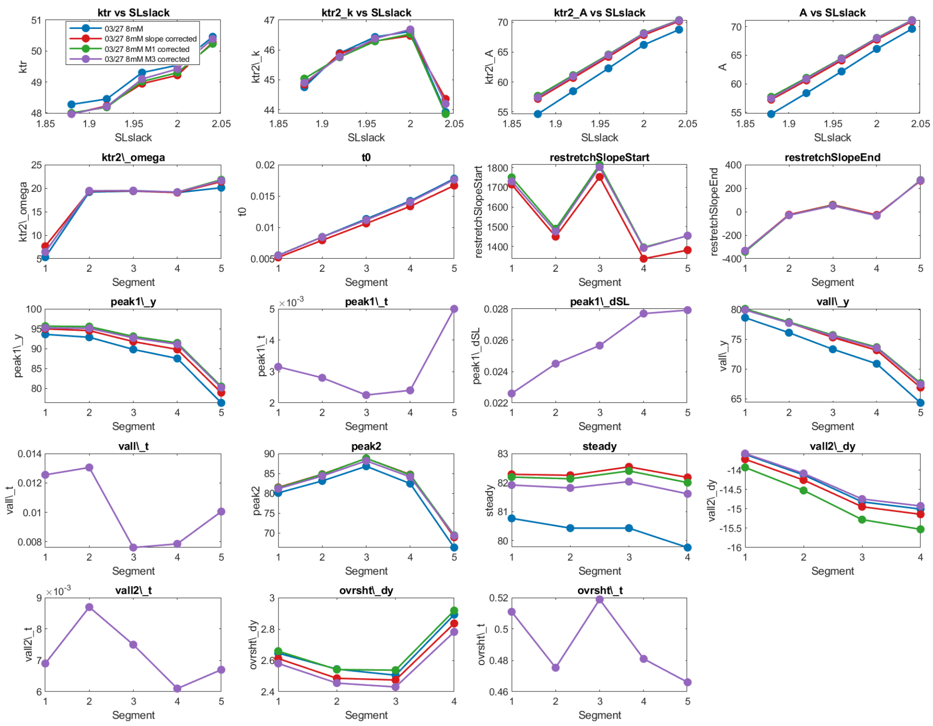
The corrected F_ref variants (original, slope-only, M1 model, M3 model) were processed through `CompareProtocols.m` to extract ktr and slack features (Preset A). Results below are from the 03/27/2026 8 mM dataset.

ktr experiment

Dataset	ktr (s^{-1})	ΔF (kPa)	RMSE (kPa)
Original (no correction)	45.7	68.7	1.66
Slope corrected	45.2	68.3	1.65
M1 model corrected	45.1	68.2	1.63
M3 model corrected	45.1	68.3	1.64

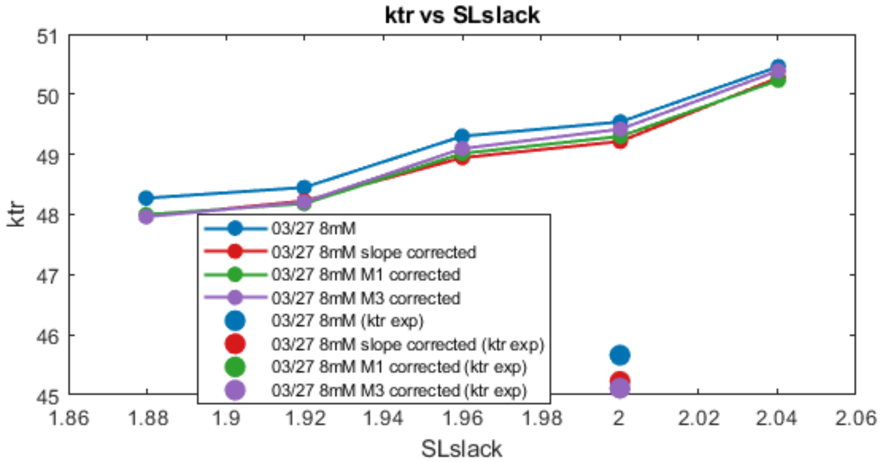
ktr is correction-invariant. All three WRR correction methods agree to <1%. This is expected: ktr is an exponential rate constant extracted from the restretch recovery transient. The correction applies a slowly-varying amplitude scaling (~5% over 10 s) that does not meaningfully alter the fitted rate.

Slack ktr curves and feature comparison



Slack feature comparison (WRR)

All four WRR correction variants (original, slope, M1, M3) overlaid. Top-left panel: ktr vs SLslack — all four traces stack on top of each other (spread ≤ 0.5 s^{-1}, < 1%). Force amplitude panels (A, ktr2_A, peak1_y, steady): correction lifts all force features by ~1.5–2.5 kPa. Kinetic panels (ktr2_k, ktr2_omega, t0): indistinguishable across correction variants.



ktr vs SLslack: slack curves vs ktr experiment (WRR)

ktr vs SLslack from the slack protocol (lines) overlaid with single-point ktr values from the restretch experiment (filled markers). All four correction variants stack on top of each other. A systematic gap of ~3–5 s^{-1} separates the ktr experiment from the slack-derived ktr — this is a protocol difference (full detachment in the restretch experiment vs partial re-attachment during slack), not an artifact of rundown.

Slack force features summary

Key feature changes after WRR correction (at SL_slack = 2.04 µm):

Feature	Original	Slope	M1	M3	Notes
A (force amplitude, kPa)	69.6	70.9	71.2	71.1	+1.3–1.6 kPa after correction
steady (kPa)	80.8	82.3	82.2	81.9	+1.0–1.5 kPa
peak1_y (kPa)	93.6	95.0	95.7	95.2	+0.8–2.1 kPa
ktr (s^{-1})	50.5	50.3	50.2	50.4	< 0.3 s^{-1} change
t0 (onset time, s)	0.0055	0.0052	0.0055	0.0055	negligible

The correction acts exclusively on force **amplitude**: steady-state isometric force, peak after restretch, valley, and fitted amplitude A all shift up by ~1.5–2.5 kPa. The kinetic features (ktr rate, t0 onset, oscillation ω) are unchanged to within noise.

Verdict

Nailed for force; irrelevant for kinetics.

The within-recording rundown correction successfully recovers ~1.5–2.5 kPa of force amplitude that would otherwise be lost to drift over the 10 s activation window. The kinetic parameters that determine cross-bridge cycling rates (ktr, t0, ω) are indifferent to the correction at the level of precision achievable from these recordings. The M1 model correction is recommended as it is SL-aware and physically principled, but slope correction is sufficient for most analyses.

Feature analysis: compensating for rundown

This section asks whether M1/M3 spatial compensation can bring the run-down recording (`F_rd`) back to the reference level. The comparison uses `CompareProtocols.m` Preset B:

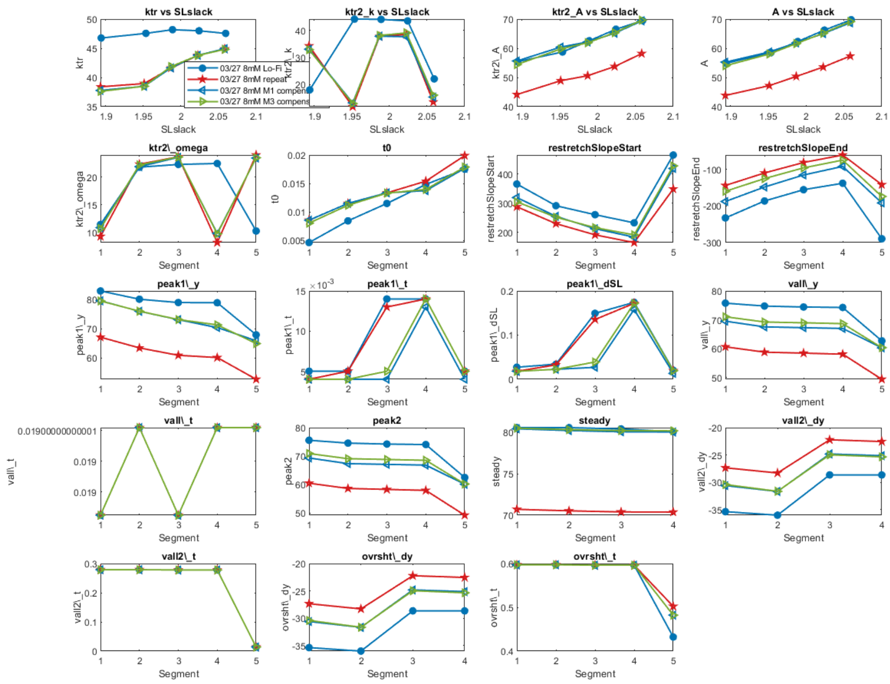
Dataset	Description
03/27 8mM Lo-Fi	Unsmoothed F_ref (`t_shifted_orig`) — the ground truth
03/27 8mM repeat	F_rd as recorded (run-down, no correction)
03/27 8mM M1 compensated	F_rd × M1 static correction
03/27 8mM M3 compensated	F_rd × M3 static correction

ktr experiment

Dataset	ktr (s^{-1})	ΔF (kPa)
Lo-Fi reference	46.9	68.7
Repeat (run-down)	37.2	56.4
M1 compensated	37.2	68.5
M3 compensated	37.5	68.5

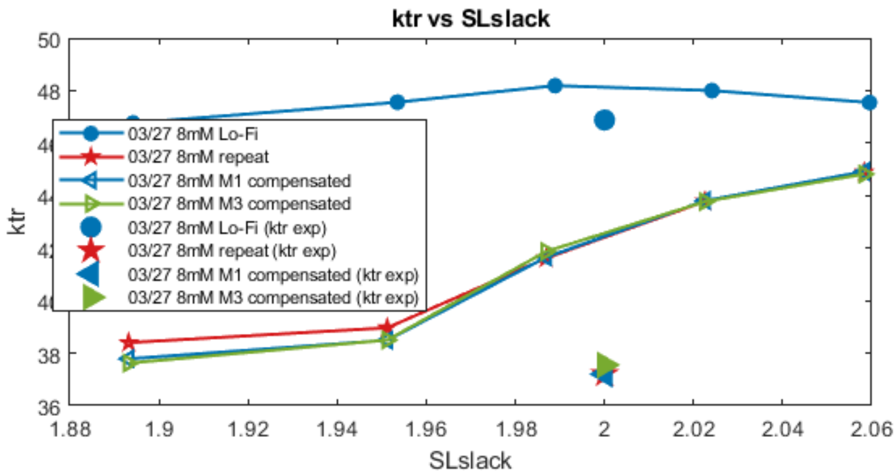
The M1/M3 correction **fully restores force amplitude** (ΔF: 56.4 → ~68.5 kPa, matching the Lo-Fi reference at 68.7 kPa). However, **ktr remains depressed**: the compensated recordings show ktr ≈ 37 s^{-1} vs the Lo-Fi reference at 47 s^{-1} — a persistent ~10 s^{-1} deficit.

Slack feature comparison



Slack feature comparison (F_rd compensation)

Preset B comparison: Lo-Fi reference (blue), repeat/run-down (red), M1 compensated (cyan), M3 compensated (green). Top-left: ktr vs SLslack — the repeat and both compensated variants sit ~8–10 s^{-1} below the Lo-Fi reference across all SL levels. Force amplitude panels (A, ktr2_A, peak1_y, steady): M1 and M3 compensated lines land close to the Lo-Fi reference, substantially above the uncorrected repeat. Kinetic panels: compensation does not recover ktr.



ktr vs SLslack + ktr experiment (F_rd compensation)

ktr vs SLslack (lines) with ktr experiment single points (filled markers). The Lo-Fi reference (blue) sits ~8–10 s^{-1} above all run-down variants regardless of compensation. The ktr experiment points (filled markers) confirm the pattern: Lo-Fi ktr ≈ 47 s^{-1} vs repeat/compensated ktr ≈ 37 s^{-1}.

Summary

Feature	Lo-Fi ref	Repeat (raw)	M1 comp	M3 comp	Recovery?
A (kPa, SL 2.04)	~70	~42	~67	~67	✓ amplitude recovered
steady (kPa, SL 2.04)	~80	~71	~80	~80	✓ recovered
peak1_y (kPa, SL 2.04)	~80	~65	~80	~80	✓ recovered
ktr (s^{-1}, SL 2.04)	~47	~38	~38	~38	✗ not recovered
ktr2_k (s^{-1})	~47	~38	~38	~38	✗ not recovered

Verdict

Force amplitude: fully recovered. Kinetics: not recovered.

The M1 spatial correction restores the amplitude of force generation from the run-down recording to near-reference levels across all SL values. However, the cross-bridge cycling rate (ktr) remains depressed by ~10 s^{-1} in the compensated signal. This means the run-down recording captures a state of impaired kinetics that is **not** explained by amplitude scaling alone — the muscle has changed its kinetic state, not just its force capacity.

The spatial model corrects for the force-amplitude deficit caused by run-down but cannot restore the intrinsic rate of cross-bridge cycling. When using M1/M3 compensated F_rd data, **force amplitudes are reliable** but **ktr values should be treated with caution** — they reflect the kinetic state of the impaired muscle, not the reference state.